

DIYA K BHAT

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PROFESSIONAL SUMMARY

AI&ML Engineer with hands-on experience building machine learning applications, LLM and RAG systems, computer vision pipelines, and production backend services. Proficient in Python, PyTorch, FastAPI, LangChain, LangGraph, PostgreSQL, Docker, and REST APIs. Experienced in model development and evaluation, semantic retrieval, real-time inference, API development, and deployment.

SKILLS

- **Programming:** Python, SQL, Java, C
- **Machine Learning & AI:** Machine Learning, Deep Learning, K-Means, LLMs, RAG, Prompt Engineering
- **Generative AI:** LangChain, LangGraph, Gemini API, Embeddings, RLHF
- **Computer Vision:** YOLOv8, OpenCV, Real-Time Object Detection
- **AI Development Tools:** Cursor, Claude, Antigravity
- **Backend & APIs:** FastAPI, Flask, Django, REST APIs, JWT Authentication
- **Data & Databases:** PostgreSQL, MySQL, MongoDB, SQLite, ETL, Data Processing, Data Modeling, Data Warehousing
- **Cloud & DevOps:** Docker, AWS, Google Cloud, Git, GitHub, Render, Netlify
- **Frontend:** React, HTML, CSS, TypeScript, JavaScript

EXPERIENCE

WizzyBox Pvt. Ltd. | AI Engineering Intern

Bengaluru, Karnataka | Sep 2025 – Jun 2026

- Engineered AI-powered LMS modules including a course recommendation engine, conversational AI Coach, automated proctoring, and real-time assessments using Python, Django, FastAPI, React, PostgreSQL, and REST APIs, supporting 500+ users.
- Developed recommendation and dynamic skill-scoring algorithms using learner profiles and outcomes, delivering personalized courses, real-time proficiency updates, and low-latency recommendations; automated assessments achieved 95% evaluation accuracy.
- Built and secured scalable backend services and JWT-authenticated REST APIs for application logic, database operations, and frontend integration, while handling testing, debugging, Docker, Git, deployment, and Agile SDLC workflows.

Outlier.ai | Freelance AI Contributor

Remote | Nov 2024 – Jan 2025

- Improved LLM output accuracy by 20% through rigorous testing, evaluation, and refinement across summarization, translation, reasoning, multilingual, and diverse content-generation tasks.
- Supported RLHF workflows by generating high-quality human preference annotations and evaluating model responses across economics, medicine, law, and psychology, contributing to more reliable and human-aligned AI systems.
- Curated and optimized structured multilingual, multi-domain datasets for AI training and evaluation pipelines, improving data quality, scalability, usability, and overall model reliability.

PROJECTS

AI Documentation Assistant using RAG | Python, FastAPI, RAG

- Built an end-to-end RAG system for querying technical documentation, implementing asynchronous ingestion, preprocessing, chunking, embedding, semantic retrieval, and grounded response generation.
- Developed an asynchronous, disk-cached ingestion pipeline with concurrent page processing and incremental updates, using all-MiniLM-L6-v2 embeddings and ChromaDB for semantic search with 800-character chunks, 150-character overlap, and top-5 retrieval.
- Integrated Gemini 1.5 Flash with context-grounding and source-citation prompts, exposing FastAPI endpoints for querying, retrieval, health checks, and indexing, with SQLite-based query logging and response-time tracking.

AI - Driven Smart Traffic Management System using Federated Approach | Python, YOLOv8, PyTorch, OpenCV, Flask

- Engineered a real-time vehicle detection pipeline with YOLOv8 inference, achieving mAP scores of 0.92 for vehicle classification and 0.96 for emergency vehicle detection.
- Designed a multi-model decision architecture that fused outputs from three YOLOv8 models using a priority-based state machine for adaptive traffic signal control.
- Processed real-time JSON traffic data, built Flask REST APIs for vehicle analytics, and developed a live multi-lane monitoring dashboard.

Spot the Job Scam | Python, Flask, React, TypeScript, XGBoost

- Built a machine learning system to detect fraudulent job postings using TF-IDF text features, SMOTE class balancing, and a stacked ensemble of XGBoost, Random Forest, and Logistic Regression.
- Achieved an F1-score of 0.84 on the fraud class, with 95% precision and 76% recall.
- Implemented real-time predictions, confidence scoring, and keyword highlighting, with the backend deployed on Render and frontend on Netlify.

A Cost-Based Restaurant Clustering and Location-Based Recommendation System | Python, K-Means, FastAPI, Gemini API

- Developed a location-based restaurant recommendation system using Python, K-Means clustering, FastAPI, HTML, and CSS, clustering restaurants by pricing and features with a 0.60 Silhouette Score.
- Integrated Google Maps API and Gemini API to deliver dynamic, location-aware recommendations for cost-effective dining options.

EDUCATION

Jyothy Institute of Technology (VTU)	Bengaluru
Bachelor of Engineering in Artificial Intelligence & Machine Learning	July 2026
CGPA: 8.8	
Daksh Gurukul - IIT Guwahati	Guwahati
Credit linked program in Data Science	May 2026

CERTIFICATIONS

- Introduction to Generative AI Learning Path - Google Cloud Skills Boost
- AWS Academy Graduate - AWS Academy Machine Learning Foundations
- Data Visualization: Empowering Business with Effective Insights simulation by Tata on Forage
- AWS Academy Graduate - AWS Academy Cloud Foundations
- Google Cloud Computing Foundations Certificate
- Gemini in Google Workspace (including Gmail, Docs, Meet, Sheets and Slides)

LEADERSHIP

- Student Placement Coordinator, coordinated placement activities between recruiters, students, and college teams.
- Python Workshop Instructor, conducted a workshop for junior students covering programming fundamentals and best practices.
- CSI Project Exhibition, led a project team that secured 2nd place.